Project Vela — Master Cookbook

v0.6.0 • beginner-friendly • multi-AI workflow • offline Windows 11 media player

What this is: the complete “how to build + how to automate” manual for Project Vela.
What’s inside: requirements (from your chat), architecture, repo map, step-by-step workflows for Cursor/Jules/ChatGPT/Gemini/GitHub, and an extensive prompt cookbook.

Important: licensing notes are an engineering checklist, not legal advice.

Source types used: (1) your requirements, (2) this repo’s code/docs, (3) official developer documentation, (4) common industry practice.

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1) What we are building (anti-drift summary)

Project Vela is a **Windows 11** desktop app for **offline** audio/video playback at conferences/live events, designed for an operator controlling playback during a show.

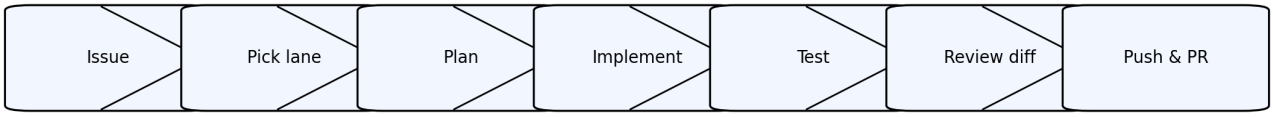
Source type: User requirements (this chat) + docs/SPEC.md (repo).

Core operator story

- Drag files in to build a playlist.
- Watch a folder (including subfolders) to auto-add new media.
- Play reliably. If something fails: output goes black, operator sees error, logs are saved.
- Hit PANIC anytime: output black plus audio mute (default).
- Cue next item in Preview window (paused on first frame, muted by default).

Source type: User requirements + docs/SPEC.md (repo).

Project Vela — Golden Loop (One Issue → One PR)



2) Non-negotiables + defaults/toggles

Rule: if any AI suggests behavior that violates these, you reject the change.

Area	Requirement	Default / toggle
Offline	Playback must work without network.	Always on (no toggle)
Stability	Never crash. On playback error: output black + operator prompt + logs.	Always on (no toggle)
Hotkeys	Hotkeys work only when app is focused.	Always on (no toggle)
Panic blackout	Hotkey + UI button; output black; audio mute default ON; leaving panic stays paused by default.	Settings toggles for audio mute + auto-resume
Playlist end	Stop at end by default; optional loop back to top.	Loop toggle in Settings

Preview window	Cue next item paused on first frame; preview muted by default.	Preview audio monitoring toggle
Folder watch	Watch local folder + subfolders; auto-add new files to playlist only.	Watch enabled toggle + include subfolders toggle
Packaging	x64 only; per-user installer; no admin required.	Always on (no toggle)

Source type: docs/SPEC.md (repo) + your chat answers (user requirements).

3) Chosen stack + licensing reality

Chosen starter stack (your Option B): **.NET + Avalonia + LibVLCSharp + Velopack**.

Source type: docs/DECISIONS.md (repo) + dependency metadata (NuGet).

Why this stack (in plain English)

- **Avalonia** gives you a modern Windows-like UI and can ship as a desktop app.
- **LibVLCSharp** wraps VLC's media engine so you get broad codec/format support (best-effort).
- **Velopack** helps produce a downloadable installer plus update packaging.

Source types: Repo decisions + official docs + package metadata.

Licensing notes (not legal advice)

LibVLC/LibVLCSharp are LGPL-licensed. Selling software can be compatible with LGPL, but you must comply with license obligations. This starter includes a licensing checklist in docs/LICENSING.md and THIRD_PARTY_NOTICES.md.

Source types: GNU LGPL license text + NuGet package license metadata + repo docs.

4) Repo map (where things live)

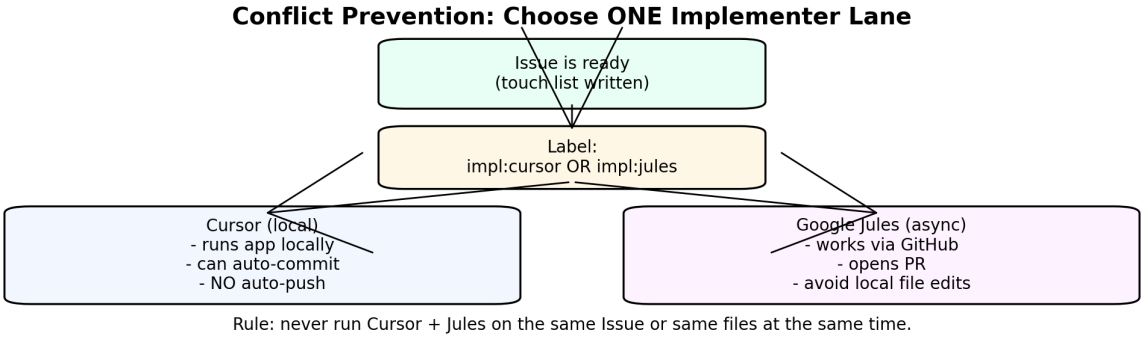
If you remember nothing else: **docs/SPEC.md is the single source of truth**.

Path	What it is	When you touch it
docs/SPEC.md	Product requirements & defaults/toggles	Before coding; when requirements change
docs/WORKFLOW_AI.md	How Cursor/Jules/ChatGPT/Gemini cooperate	When workflow drifts
docs/PROMPTS.md	Copy/paste prompts for all AIs	Every new Issue
src/ConferencePlayer.App	Avalonia UI + ViewModels	UI changes and operator workflows
src/ConferencePlayer.Playback	IPlaybackEngine + LibVLCSharp implementation	Playback behavior
src/ConferencePlayer.Core	Settings, playlist model, folder watch, logging	Stability + data model
.github/workflows	CI + release automation	When builds/releases fail
.cursor/rules	Cursor guardrails	When Cursor starts hallucinating
AGENTS.md	Instructions that Jules reads	When Jules needs better guidance

Source type: Repo artifact (file layout).

5) Your error-proof workflow (exact order)

This is the “do it the same way every time” loop. The goal is: automation without chaos.



A. Write the Issue (GitHub)

- Goal / Non-goals / Defaults & toggles / Acceptance criteria
- Touch list (files allowed to change)
- Pick one implementer lane label

B. Plan (ChatGPT)

- Ask for plan + risk list + verification steps
- Ask it to quote docs/SPEC.md headings (prevents drift)

C. Red-team (Gemini)

- Ask Gemini to attack the plan: crash paths, monitor changes, codec edge cases
- Add missing acceptance checks back into the Issue

D. Implement (Cursor OR Jules)

- Cursor: local iteration; run tests; commit; stop; no push
- Jules: run task from Issue; review plan; let it open PR

E. Verify (You)

- Smoke test (docs/SMOKE_TEST.md)
- Glance at diff vs Issue + touch list
- Push (Cursor lane) and open PR

F. Merge + release

- Merge only when CI is green
- Use Velopack workflow to build installer artifacts

Source types: Common industry practice + official docs + your chosen workflow preference.

6) Tool setup (beginner friendly)

6.1 GitHub setup

- Enable protected branch rules for **main**: require PR, require status checks (CI).
- Use Issues as the source of truth: every change starts with an Issue.
- Use labels: impl:cursor OR impl:jules (never both).

Link: GitHub protected branches docs.

Source type: GitHub official documentation + repo conventions.

6.2 Cursor setup (local agent)

- Open the repo in Cursor and confirm **.cursor/rules** exists.
- Enable agent terminal only if you will review executed commands.
- Let Cursor commit to your branch; you push after reviewing the diff.

Links: Cursor rules • Cursor agent terminal • Cursor security.

Source types: Cursor official documentation + your workflow preference.

6.3 Google Jules setup (async PR agent)

- Jules reads **AGENTS.md** at repo root for tool conventions.
- Start a task from a GitHub Issue by applying the label **jules**.
- Jules runs tasks in a VM that clones your repo and installs dependencies.

Links: Jules docs • Running tasks.

Source type: Jules official documentation.

6.4 ChatGPT setup (planner + reviewer)

- Create a ChatGPT **Project** named “Project Vela”.
- Upload repo docs so ChatGPT can quote them (SPEC, WORKFLOW, PROMPTS).
- Put the Universal Header into your Project instructions.

Links: Projects in ChatGPT • File uploads FAQ.

Source type: OpenAI official documentation.

6.5 Gemini Ultra setup (red-team)

- Use Gemini for red-team thinking: how could this crash or fail in production?
- Ask for offline experiments: CPU usage, long-play stability, monitor disconnect.

Link: Gemini thinking documentation.

Source types: Google official documentation + common industry practice.

7) Conflict prevention protocol (Cursor vs Jules)

Goal: you should almost never see a merge conflict. Conflicts are usually a workflow issue, not a technical requirement.

The rule set

- **One implementer per Issue:** impl:cursor OR impl:jules.
- **Touch list required** before implementation starts.
- **File ownership:** the agent who owns the Issue is the only one editing those files.
- **Cursor commits, human pushes:** reduces accidental remote pollution.
- **If you must switch lanes:** close/merge the current PR first, then relabel the Issue and update touch list.

Source types: Common industry practice + your chosen workflow preference.

Emergency procedure: I accidentally ran both AIs

- Stop one agent immediately.
- Pick which branch/PR to keep.
- Cherry-pick only the necessary commits onto a clean branch.
- Update the Issue with the new touch list and lane label.

Source type: Common industry practice (git conflict recovery).

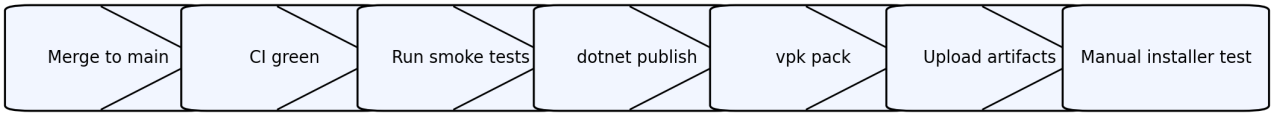
8) Automation (CI + packaging + release)

Automation makes you faster, but only if it's predictable. This repo ships with:

- **CI**: dotnet restore/build/test on PRs and pushes.
- **Release workflow**: packages a Velopack release artifact on demand.
- **Lane guard**: warns if both impl:cursor and impl:jules labels are set.

Source type: Repo artifact (.github/workflows).

Release pipeline



Velopack notes (per-user, no privileged directories)

Velopack's Windows packaging documentation describes installation under the user profile (not system folders) and includes a section on preserving user files and settings across updates.

Links: Windows packaging • Preserving files.

Source type: Velopack official documentation.

9) Prompt cookbook (copy/paste library + summaries)

These prompts are designed to be beginner-safe: they force plans, touch lists, tests, and admissions of uncertainty.

Critical rule: if you do not paste the Issue + acceptance criteria + touch list, the AI will guess. Always include them.

Prompt index (summary for each)

Tool	Prompt name	When to use	Expected output
ChatGPT	Issue drafting	Before coding	Issue text + acceptance criteria
ChatGPT	Lane decision	Start of every Issue	impl:cursor vs impl:jules + touch list
ChatGPT	Mini design note	Playback/monitor/file-watch work	Approach + failure modes
ChatGPT	Beginner test plan	Before implementation	Unit + smoke checklist
ChatGPT	PR review	Before merge	Blockers + missing tests
Cursor	Implement Issue (commit/no push)	impl:cursor Issues	Code + tests + local commit
Cursor	Bug fix from logs	When you have a log snippet	Root cause + smallest fix
Cursor	Add setting/toggle	Any new option	UI + persistence + docs update
Cursor	Add hotkey	New operator control	Focused-only hotkey + smoke test
Jules	Implement Issue (small PR)	impl:jules Issues	Async PR + verification steps
Jules	Docs only	Safe background cleanup	Docs PR only
Jules	Tests only	Increase coverage	Tests PR only
Gemini	Red-team plan	After planning	Failure modes + mitigations
Gemini	Break the release checklist	Before shipping	20 stress tests + expected behavior

Source type: Common industry practice + this repo's workflow docs.

Copy/paste prompts

ChatGPT — Lane decision + touch list

```
You are working on Project Vela.

Inputs:
- GitHub Issue: <PASTE>
- Relevant spec excerpt (docs/SPEC.md): <PASTE>

Task:
1) Choose ONE lane: impl:cursor OR impl:jules (with reasoning).
2) Produce a touch list (files allowed to change).
3) Produce a 6-10 step plan.
4) Produce a verification plan (unit + smoke).
5) If anything is unknown, say: "I am uncertain about this" and ask for the missing info.
```

Cursor — Implement Issue (commit allowed, no push)

```
Implement Issue #<N> locally.

Constraints:
- Respect docs/SPEC.md and docs/ARCHITECTURE.md.
- No new dependencies unless explicitly allowed.
- Keep changes small.
- You may commit, but DO NOT push unless I ask.

Output:
- Summary
- Files changed
- Commands run
- Smoke test steps
```

Cursor — Bug fix from logs (smallest fix first)

```
Fix a bug using this information.

Repro steps:
<PASTE>

Log snippet:
<PASTE>

Requirements:
- Never crash posture: output black + operator prompt + logs.
- Add or improve logging so future failures are diagnosable.
- Keep the fix minimal.

Output:
- Root cause hypothesis (file + method)
- Smallest safe fix
```

```
- Tests to add/update
```

Jules — Implement Issue (small PR)

```
Implement GitHub Issue #<N>.

Constraints:
- Respect the Issue touch list.
- Keep PR small (~300 LOC).
- No new dependencies unless explicitly permitted.
- Update docs/SMOKE_TEST.md if behavior changes.
- Add/update tests where possible.

Deliverable:
- PR with a clear summary + verification steps.
```

Gemini — Red-team the plan

```
Act as a red-team reviewer for Project Vela.

Given this plan:
<PASTE>

Attack it for:
- crash risks
- file watch pitfalls
- multi-monitor pitfalls
- operator workflow hazards

Output:
- Failure modes
- Mitigations
- Missing acceptance checks
```

Gemini — Break the release checklist

```
Act as QA lead for an offline Windows media player.

Given features:
- panic blackout
- preview cue next
- folder watch (subfolders)
- playlists + loop toggle

Create 20 short stress tests:
- steps
- expected result
- what to check in logs
```

Full prompt library lives in **docs/PROMPTS.md** (in the zip).

10) Troubleshooting + what to do when AI goes weird

Most failures are predictable. Here are the patterns.

AI changed too many files

- Close PR, revert, rewrite Issue with a tighter touch list.
- Ask ChatGPT to split into smaller Issues.

AI added a new dependency

- Stop. Ask: do we truly need it?
- If no, revert. If yes, add license review to the Issue.

Playback error handling missing

- Reject the PR. The app must never crash.
- Add operator-visible error + logs + output black.

Cursor pushed junk to GitHub

- Revert remote commit, enforce: commit then diff review then push.

Jules and Cursor both edited the same file

- Pick one PR; cherry-pick clean commits; close the other.
- Re-label Issues and re-assign lanes.

Source type: Common industry practice.

11) Open questions (things to decide soon)

- Do you want code signing for the installer (recommended for Windows SmartScreen reputation)?
- Do you want an auto-update channel (Velopack can do updates, but you can ship offline-only first)?
- Do you want a standardized set of sample media files for regression testing? (Even 3–5 files helps.)
- Do you want crash reporting beyond local logs? (This can conflict with offline-only unless optional.)

Source type: Common industry practice + unresolved product decisions.

12) Sources (clickable)

Item	Link	Source type
Cursor docs — Rules	https://cursor.com/docs/context/rules	Official developer documentation
Cursor docs — Agent terminal	https://cursor.com/docs/agent/terminal	Official developer documentation
Cursor security	https://cursor.com/security	Official developer documentation
Jules docs — Getting started	https://jules.google/docs/	Official developer documentation
Jules docs — Running tasks (includes Issue label “jules”)	https://jules.google/docs/running-tasks/	Official developer documentation
GitHub docs — Protected branches	https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/managing-protected-branches/about-protected-branches	Official developer documentation
Velopack docs — Packaging overview	https://docs.velopack.io/packaging/overview	Official developer documentation
Velopack docs — Windows packaging	https://docs.velopack.io/packaging/operating-systems/windows	Official developer documentation
Velopack docs — Preserving files and settings	https://docs.velopack.io/packaging/preserving-files	Official developer documentation
NuGet — LibVLCSharp.Avalonia package (license metadata)	https://www.nuget.org/packages/LibVLCSharp.Avalonia	Official package registry metadata
NuGet — VideoLAN.LibVLC.Windows package (bundled native binaries)	https://www.nuget.org/packages/VideoLAN.LibVLC.Windows	Official package registry metadata
GNU — LGPL v2.1 license	https://www.gnu.org/licenses/old-licenses/lgpl-2.1.en.html	License text (authoritative)
OpenAI Help — Projects in ChatGPT	https://help.openai.com/en/articles/10169521-projects-in-chatgpt	Official developer documentation

OpenAI Help — File uploads FAQ	https://help.openai.com/en/articles/8555545-file-uploads-faq	Official developer documentation
Google AI Dev — Gemini thinking	https://ai.google.dev/gemini-api/docs/thinking	Official developer documentation